SAP - SERTICA API Summary

Required request payloads for master data, stock receipt, and consumption polling

0. Authentication

Obtain a separate bearer token for every SERTICA unit context used by the integration.

Request

```
POST /api/Auth
Content-Type: application/json

{
"login": "<integration-user>",
"password": "<secret>",
"roleNo": "<authorized-role>",
"unitNo": "Master Unit"
}
```

Required fields

Field	Meaning
login	SERTICA integration login.
password	Stored securely; never embedded in source code or logs.
roleNo	Authorized SERTICA role, for example Admin during development.
unitNo	Authentication context: Master Unit, KPG0, IHB0, etc.

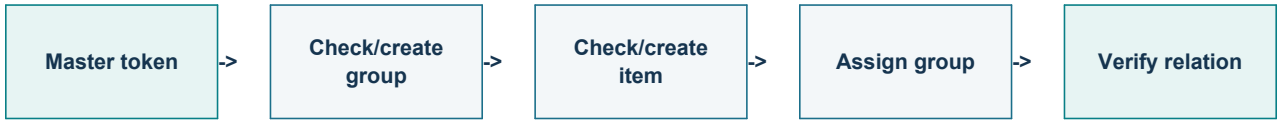
Token usage

Token context	Use
Master Unit	Create/update master item groups, items, and item-group membership.
Vessel	Distribute groups, verify visibility, assign warehouse/location, adjust stock, and read vessel consumption.

Response

Read `accessToken` and `expiresIn`. Send `Authorization: Bearer <token>` on later calls. A token does not change unit context when a request field changes.

1. Create Master Group and Item



A. Create the group only when missing

```
GET /api/ItemGroups/<parentGroupNo>
GET /api/ItemGroups/<groupNo>
POST /api/ItemGroups
Authorization: Bearer <master-unit-token>

{
"dataOwner": "MasterUnit",
"itemGroupNo": "<SAP group code>",
"name": "<SAP group name>",
"text": "Synchronized from SAP",
"parentItemGroupNo": "<approved parent>",
"unitNo": "Master Unit"
}
```

B. Create the MasterUnit item

```
GET /api/Items/<itemNo>
POST /api/Items
Authorization: Bearer <master-unit-token>

{
"dataOwner": "MasterUnit",
"itemNo": "<SAP material number>",
"name": "<SAP material description>",
"measureUnitNo": "PCS",
"itemTypeNo": "SPARE",
"currencyNo": "IDR",
"unitNo": "Master Unit",
"inactive": false,
"critical": false,
"certificate": false,
"trackExpiry": false,
"trackedItem": false,
"disallowManualStockAdjustment": false
}
```

Create rule

Use GET before POST. If the business key already exists, compare ownership and update with PUT rather than creating a duplicate.

1. Assign and Distribute Master Item

C. Assign the group to the item

```
GET /api/ItemGroups/<groupNo>
GET /api/Items/<itemNo>
PUT /api/Items/<itemNo>
Authorization: Bearer <master-unit-token>

Merge into the latest item document:
"itemGroups": [{
"itemNo": "<itemNo>",
"itemName": "<item name>",
"itemGroupNo": "<groupNo>",
"itemGroupName": "<group name>",
"itemGroupUnitNo": "Master Unit",
"itemGroupUnitName": "Office",
"guidForReferencedObject": "<group.guid>",
"unitGuid": "<group.unitGuid>",
"deleted": false
}]
Verify: GET /api/Items/<itemNo>/ItemGroups
```

D. Distribute to one vessel

```
GET /api/ItemGroupSelection/<groupNo>
Authorization: Bearer <target-vessel-token>

PUT /api/ItemGroupSelectionTree
Authorization: Bearer <target-vessel-token>

[
{
"dataOwner": "Unit",
"itemGroupGuid": "<selection.itemGroupGuid>",
"hasChildren": false,
"name": "<group name>",
"itemGroupText": "<group text>",
"itemGroupNo": "<groupNo>",
"parentVisibleKey": "<parentGroupNo>",
"unitNo": "Master Unit",
"selected": true,
"selectedUnitNo": "<target vessel code>",
"hasParent": true
}
]
```

E. Assign vessel warehouse/location

```
GET /api/Items/<itemNo>
PUT /api/Items/<itemNo>
Authorization: Bearer <target-vessel-token>

Merge into the latest item document:
"warehouses": [{
"warehouseNo": "E01", "warehouseName": "Engine Room",
"locationNo": "001.001", "locationName": "BIN 01",
"standardWarehouse": true, "stock": 0,
"maximumStock": 0, "minimumStock": 0,
"itemNo": "<itemNo>", "itemName": "<item name>",
"unitNo": "KPG0", "itemUnitNo": "Master Unit",
"measureUnitNo": "PCS"
}]
Verify: GET /api/Items/<itemNo>/warehouses
```

2. Goods Receipt - Increase SERTICA Stock

Authenticate as the receiving vessel. Resolve and verify the item warehouse/location before posting.



A. Delta receipt payload

```
POST /api/Items/<itemNo>/adjuststock
Authorization: Bearer <receiving-vessel-token>
Content-Type: application/json

{
"warehouseNo": "E01",
"itemLocationNo": "001.001",
"quantity": 5,
"description": "SAP_GR:<unique-reference>",
"stockAdjustmentType": "WriteUpStock",
"fromSerticaApp": false,
"sourceSystem": "SAP"
}
```

B. Absolute balance payload

```
{
"warehouseNo": "E01",
"itemLocationNo": "001.001",
"quantity": 5,
"description": "SAP_STOCKTAKE:<unique-reference>",
"stockAdjustmentType": "Stocktake",
"fromSerticaApp": false,
"sourceSystem": "SAP"
}
```

Required fields

Field	Rule
warehouseNo	Receiving vessel warehouse.
itemLocationNo	Omit only when the warehouse assignment has no location.
quantity	Receipt delta for WriteUpStock; target balance for Stocktake.
description	Unique SAP correlation reference used for audit and echo suppression.
stockAdjustmentType	WriteUpStock for receipt delta; Stocktake for absolute reconciliation.

Verification status

Stocktake was directly tested and read back successfully. WriteUpStock is exposed by the schema but the direct request still requires a dedicated development test before production.

3. Consumption Option A - Job Consumption

SERTICA -> SAP. Use the vessel token and poll finalized job executions.



A. Search payload

```
POST /api/JobHistories/search?page=1&pageSize=100
Authorization: Bearer <vessel-token>
Content-Type: application/json

{
"distinct": true,
"columns": [
"guid", "jobNo", "liveJobNo", "status",
"linkDate", "unitNo", "updateDate", "createDate"
],
"criteria": [
{"field": "unitNo", "operator": "Equals", "values": ["KPG0"]},
{"field": "status", "operator": "Equals", "values": ["Final"]}
]
}
```

B. Read execution and item lines

```
GET /api/JobHistories/<jobHistoryGuid>
GET /api/JobHistories/<jobHistoryGuid>/items
Authorization: Bearer <vessel-token>
```

Fields required for SAP consumption

Field	Use
status	Process Final executions only.
quantityFromOwnStock	Inventory quantity consumed from vessel stock.
itemNo	SAP material mapping.
warehouseNo / itemLocationNo	SAP plant/storage-location mapping.
jobHistoryGuid + item-line guid	Durable deduplication and SAP external reference.

Deduplication key

`SERTICA_JOB:<unitNo>:<jobHistoryGuid>:<jobHistoryItemGuid>`. Do not also post its WarehouseHistory row as a second SAP goods issue.

4. Consumption Option B - Warehouse History

SERTICA -> SAP. This stream includes job use, manual adjustments, initialization, receipts, and other stock movements, so classification is mandatory.

Search payload

```
POST /api/ItemWarehouseHistories/search?page=1&pageSize=100
Authorization: Bearer <vessel-token>
Content-Type: application/json

{
"distinct": true,
"columns": [
"guid", "itemNo", "warehouseNo", "locationNo",
"quantity", "historyType", "historyDate",
"stockQuantityAfterTransaction", "description",
"updateUserNo", "unitNo"
],
"criteria": [
{"field": "unitNo", "operator": "Equals", "values": ["KPG0"]},
{"field": "historyDate", "operator": "GreaterThan",
"values": ["<checkpoint-date-time>"]}
]
}
```

Classification

Warehouse event	SAP handling
JobHistory	Exclude if Option A is active; it represents the same job consumption.
Init	Do not post as consumption.
Approved manual negative adjustment	Post as consumption or inventory difference using an approved type/reason mapping.
Positive adjustment / return	Not consumption; handle as a separate receipt or return flow if in scope.
Transfer / reversal / correction	Map explicitly; never classify using the quantity sign alone.
SAP_GR / SAP_SYNC description	Suppress the outbound echo of an SAP-originated stock update.

Deduplication key

`SERTICA_STOCK:<unitNo>:<warehouseHistoryGuid>`. Process every page and keep a per-vessel checkpoint with overlap.

Production check

The warehouse-history search shape is verified. Manual WriteDownStock consumption, date-filter syntax, timezone behavior, and reversal handling still require end-to-end validation.